

ENTERPRISE PRODUCT REQUIREMENTS DOCUMENT (PRD)

# VQube Industrial Engineering Platform & Global Client Delivery Ecosystem

A Comprehensive Engineering Specification for Multi-Disciplinary Plant Consulting, Numerical Simulation (CFD/FEA), Rapid Manpower Deputation, and High-Performance Digital Operations.

|                       |                                                                |                      |                                                            |
|-----------------------|----------------------------------------------------------------|----------------------|------------------------------------------------------------|
| Document Reference    | PRD-VQT-2026-V2.4                                              | Release Status       | APPROVED & BASELINED                                       |
| Document Author       | Senior Technical Project & Product Manager                     | Author Function      | Engineering Strategy & Digital Platform Operations         |
| Executive Sponsor     | Vadivelu Dhamodharan                                           | Sponsor Title        | Head — Engineering & Business Solutions                    |
| Target Industries     | Cement, Mining & Minerals, Oil & Gas, Power, Heavy EPC         | Governing Codes      | ASME Section VIII, B31.1/B31.3, API 650/620, DIN, ISO 9001 |
| Digital Platform Spec | Native HTML5/CSS3/ES6+, PWA Service Worker, /lms.txt Compliant | Current Version Date | October 2026                                               |

**Executive Project Summary:** This Product Requirements Document establishes the technical, operational, and commercial architecture for **VQube Technologies [VQT]**. It codifies the dual-pillar ecosystem encompassing: (1) Core Industrial Plant Engineering, Advanced Simulation (CFD/FEA/CAESAR II), and Global Engineering Manpower Deputation across 11 technical specialisms; and (2) The Digital Client Engagement Infrastructure, featuring sub-50ms repeat load PWA resilience, enterprise SEO, structured JSON-LD knowledge graphs, and machine-readable autonomous procurement endpoints (/lms.txt).

## 0 Document Governance & Executive Sign-Off

This document serves as the single source of truth (SSOT) governing the design, execution, verification, and commercial delivery of VQube Technologies' engineering services and digital platform. Any deviations or change requests must proceed through the formal Engineering Change Control Board (ECCB).

### 0.1 Document Revision History

| Rev  | Date        | Author / Role                | Description of Changes                                                                                                            | Status     |
|------|-------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|
| v1.0 | 15 Jan 2025 | Senior PM & Systems Eng      | Initial release establishing Cement & Mining Plant Engineering baseline and GA platework deliverables.                            | ARCHIVED   |
| v1.5 | 10 Jun 2025 | Senior Technical PM          | Added Oil & Gas piping, ASME Section VIII pressure vessels, API 650/620 storage tanks, and CAESAR II stress analysis.             | ARCHIVED   |
| v2.0 | 05 Jan 2026 | Senior Product Manager       | Introduced 11-Discipline Engineering Manpower Deputation, GDC offshore delivery center, and SLA benchmarks (5-14 days).           | ARCHIVED   |
| v2.2 | 14 May 2026 | Lead Digital Architect       | Integrated PWA Service Worker caching architecture (Cache-First + Stale-While-Revalidate) for sub-50ms repeat loads.              | SUPERSEDED |
| v2.4 | 27 Sep 2026 | Senior Project & Product Mgr | Enterprise Baselined PRD: Machine-Readable AI endpoints (/llms.txt), Schema.org JSON-LD graph, and Milestone Stage-Gate controls. | BASELINED  |

### 0.2 Multi-Disciplinary Stakeholder Sign-Off Matrix

| Stakeholder Role       | Name & Title                     | Signature / Authorization      | Date & Approval Status |
|------------------------|----------------------------------|--------------------------------|------------------------|
| Project & Product Lead | Senior Technical Project Manager | [Signed Digitally - PMO]       | 27 Sep 2026 • APPROVED |
| Head of Engineering    | Vadivelu Dhamodharan             | [Authorized - Exec Leadership] | 27 Sep 2026 • APPROVED |
| QA/QC & Code Inspector | Lead QA/QC & Welding Inspector   | [Signed Digitally - QA]        | 27 Sep 2026 • APPROVED |
| Commercial Operations  | Head of Commercial & Contracts   | [Signed Digitally - Contracts] | 27 Sep 2026 • APPROVED |

### 0.3 Table of Contents Overview

|                                                                |                                                                 |
|----------------------------------------------------------------|-----------------------------------------------------------------|
| Section 1: Executive Summary & Strategic Context               | Section 6: Non-Functional Requirements (NFR-1 to NFR-5)         |
| Section 2: Product Vision, Goals & Success Metrics (OKRs/KPIs) | Section 7: Commercial Engagement Models & Milestone Stage-Gates |
| Section 3: Target Personas & Enterprise Stakeholder Journeys   | Section 8: Implementation Roadmap, Phasing & RACI Matrix        |
| Section 4: System Architecture & Dual-Pillar Framework         | Section 9: Risk Management, Mitigation & Contingency Plan       |
| Section 5: Detailed Functional Requirements (FR-1 to FR-6)     | Section 10: Verification, Acceptance & Definition of Done       |



## 1 Executive Summary & Strategic Context

50+

Delivered Global Projects

15+

Enterprise EPC Clients

100%

Code-Compliant Design

5-14d

Manpower Mobilization SLA

### 1.1 Enterprise Background & Core Mandate

**VQube Technologies [VQT]** is an international engineering consultancy and technical workforce solutions provider operating out of its Global Delivery Center in Chennai, India. Rooted in the corporate philosophy of "*Vector • Vision • Value*", VQube delivers end-to-end multi-disciplinary engineering, high-fidelity numerical simulation, and skilled engineering manpower deputation to heavy process industries worldwide.

### 1.2 Market Problem Statement & Heavy Industry Friction

Heavy manufacturing and process engineering sectors (Cement, Mining & Minerals, Oil & Gas, Power Generation) face severe operational and project delivery challenges:

- **Late-Stage Fabrication & Erection Failures:** Unidentified thermal expansion stresses, aerodynamic pressure drops, and structural resonance cause multimillion-dollar field modifications and commissioning delays.
- **Shortage of Multi-Disciplinary Engineering Talent:** EPC contractors frequently face crippling shortages of certified piping stress engineers (CAESAR II), Tekla steel modelers, process CFD specialists, and licensed QA/QC inspectors.
- **Fragmented Engineering Hand-offs:** Conceptual design, detailed 3D modeling, stress verification, and procurement packages are often executed by disconnected vendors, leading to clash issues and schedule slippage.
- **Digital Discovery & Procurement Inefficiency:** Traditional engineering consultancies lack transparent, machine-readable specifications, preventing autonomous enterprise evaluation and rapid technical scoping.

### 1.3 The VQube Solution & Unique Value Proposition

VQube resolves these bottlenecks through an integrated, closed-loop delivery ecosystem combining:

- **Pre-Fabrication Multi-Physics Verification:** Leveraging ANSYS Fluent (CFD), ANSYS Mechanical (FEA), and CAESAR II to eliminate duct turbulence, burner instability, and thermal stress cracks prior to IFC release.
- **Rapid Deployment Across 11 Engineering Disciplines:** Pre-screened talent mobilized within 5 to 14 business days across onsite plant assignments or offshore dedicated teams.
- **Strict Stage-Gate Quality Control:** Milestone-based engineering reviews at 30%, 60%, 90%, and 100% IFC issue ensure absolute compliance with ASME, ASTM, DIN, ISO, and API codes.
- **Ultra-High-Performance Digital Portal:** Built on native semantic web standards with PWA offline caching, structured Schema.org graphs, and the emerging /11ms.txt standard for next-generation automated sourcing.

#### The 6 Operating Pillars of VQube:

- 01. **Trust & Integrity:** Uncompromising technical transparency and milestone accountability.
- 02. **Quality & Excellence:** 100% code-compliant engineering verified by advanced numerical modeling.
- 03. **Creativity & Innovation:** Continuous optimization of plant aerodynamics, steel weight, and flow efficiency.
- 04. **Partnership & Customer Focus:** Flexible commercial engagement tailored to capital budget constraints.
- 05. **Teamwork & Collaboration:** Seamless integration with client owner teams, EPCs, and equipment OEMs.
- 06. **Safety & Sustainability:** Inherently safe designs, lower aerodynamic pressure losses, and reduced carbon intensity.

2 Product Vision, Goals & Success Metrics (OKRs & KPIs)

2.1 Strategic North Star & Objective Key Results (OKRs)

The North Star Metric: "Zero-Defect Engineering Delivery Rate (ZDER) • Rapid Mobilization SLA (≤ 10 Days) • Sub-50ms Client Interface Latency."

| Objective                                                                          | Key Results (Target FY 2026-2027)                                                                                                                                                                                                                                          | Cadence   | Owner                    |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------|
| <b>Objective 1:</b><br>Establish Global Engineering Consulting Dominance           | <ul style="list-style-type: none"><li>Deliver 30+ major EPC project packages with 0% code non-compliance.</li><li>Expand client base across GCC, Southeast Asia, Europe, and Americas.</li><li>Achieve ≥ 90% repeat business and client retention rate.</li></ul>          | Quarterly | Head of Engineering      |
| <b>Objective 2:</b><br>Scale Engineering Manpower Deputation                       | <ul style="list-style-type: none"><li>Maintain talent pool of 250+ vetted engineers across 11 disciplines.</li><li>Reduce average client onboarding timeline to 5–10 business days.</li><li>Achieve 98%+ client satisfaction score on deputed talent.</li></ul>            | Monthly   | Talent & Deputation Lead |
| <b>Objective 3:</b><br>Lead Heavy Industry in Digital Performance & AI Integration | <ul style="list-style-type: none"><li>Maintain 100/100 Core Web Vitals and sub-50ms repeat page loads.</li><li>Provide 100% compliant /llms.txt endpoints for automated RFQ bots.</li><li>Achieve top 3 Google organic rankings for industrial FEA/CFD keywords.</li></ul> | Bi-Weekly | Senior Digital PM        |

2.2 Operational, Commercial & Technical KPI Matrix

| Metric / KPI                   | Current Baseline | Target Threshold | Measurement Methodology                                             | Impact Classification |
|--------------------------------|------------------|------------------|---------------------------------------------------------------------|-----------------------|
| First-Pass IFC Acceptance      | 94.2%            | ≥ 98.5%          | Audit of client review comments during Stage 4 sign-off.            | CRITICAL QUALITY      |
| Simulation SLA Turnaround      | 18 Days          | 7 – 14 Days      | Elapsed days from CAD receipt to final technical simulation report. | OPERATIONAL SLA       |
| Manpower Mobilization SLA      | 12 Days          | 5 – 10 Days      | Elapsed time from MSA signing to resource plant check-in.           | COMMERCIAL SLA        |
| Largest Contentful Paint (LCP) | 0.95s            | < 0.80s          | Chrome UX Report (CrUX) and Lighthouse v12 performance audits.      | DIGITAL WEB           |
| Repeat Load Latency (PWA)      | 35ms             | < 25ms           | Service Worker Cache-First response timing under 3G emulation.      | DIGITAL WEB           |
| Code Compliance Pass Rate      | 100%             | 100% Mandatory   | Third-party audit against ASME, DIN, ISO, and API standards.        | STATUTORY ZERO-DEFECT |

### 3 Target Enterprise Personas & Stakeholder Journeys

**Persona 1: EPC Project Director / Procurement Head**  
*"Rajesh Menon • Global Tier-1 EPC Contractor • Greenfield Plants"*

- Needs:** Reliable engineering partner for complete plant layout, FEED packages, ductwork fabrication drawings, and Tekla detailing.
- Pain Points:** Budget overruns due to steel tonnage excess; clash errors discovered on site; vendor missed deadlines.
- Success Criteria:** Milestone-based fixed-price EPCM model, 100% ASME/DIN compliance, and clash-free IFC drawing packages.

**Persona 2: Plant Technical & Maintenance Director**  
*"Dr. Hans-Ulrich Becker • Cement & Mineral Processing Conglomerate"*

- Needs:** Root-cause failure analysis for kiln refractory thermal stress, cyclone preheater clogging, and transfer chute wear.
- Pain Points:** Unplanned kiln outages costing \$150k/day; excessive aerodynamic pressure drops causing high fan power consumption.
- Success Criteria:** High-fidelity ANSYS Fluent/FEA diagnostics within 7-14 days with actionable redesigns.

**Persona 3: Global Resource Mobilization Lead**  
*"Sarah Jenkins • VP Human Resources & Project Staffing"*

- Needs:** Rapid deployment of certified piping stress engineers, Tekla modelers, and QA/QC inspectors for overseas plants.
- Pain Points:** 60-day agency recruitment lags; candidates failing technical screening during commissioning.
- Success Criteria:** Fully vetted, multi-disciplinary technical personnel mobilized within 5 to 14 business days.

**Persona 4: Autonomous AI Procurement Bot / Evaluator**  
*"Enterprise LLM Agent • Autonomous Vendor Sourcing"*

- Needs:** Structured, deterministic, machine-readable capability data, pricing models, and compliance certifications.
- Pain Points:** Unstructured marketing buzzwords; heavy client-side JavaScript rendering blocking content extraction.
- Success Criteria:** Clean Markdown endpoints (/llms.txt, /llms-full.txt) and Schema.org JSON-LD graph.

#### 3.2 End-to-End Enterprise Stakeholder Journey Map



## 4 System Architecture & Dual-Pillar Product Scope

The VQube Technologies ecosystem operates on a **Dual-Pillar Architecture** that harmonizes industrial engineering operations with a high-velocity digital engagement platform:

**PILLAR I: INDUSTRIAL ENGINEERING & SIMULATION SUITE**  
**Domain Delivery Engines:** Cement Plant Engineering • Mining & Mineral Beneficiation • Upstream & Downstream Oil & Gas • Heavy Platework.  
**Simulation Toolchain:** ANSYS Fluent (CFD) • ANSYS Mechanical (FEA) • CAESAR II Pipe Stress • AVEVA E3D • Tekla Structures.  
**Manpower Deputation:** 11 Technical Specialisms • Onsite Global • Chennai Offshore GDC • Hybrid Model.

↕ **INTEGRATED ENTERPRISE INTERFACE** ↕

**PILLAR II: HIGH-PERFORMANCE DIGITAL WEB & AI PLATFORM**  
**Core Web Tier:** Semantic HTML5 • Vanilla CSS3 Tokens • ES6+ JavaScript • Sub-50ms Repeat Loads.  
**Service Worker (sw.js):** Cache-First for static assets • Stale-While-Revalidate for HTML • Offline Resilience.  
**Autonomous AI Endpoints:** /llms.txt • /llms-full.txt • Schema.org JSON-LD Knowledge Graph (Organization, FAQ, Person).  
**Commercial Gateways:** Interactive Consultation Modal • Direct WhatsApp API • Form Validation & CRM Bridge.

### 4.2 Computational Toolchain & Governing Standards Matrix

| Discipline                         | Software Stack                  | Governing Statutory Standards                          | Typical Deliverables & Output Artifacts                                                                                            |
|------------------------------------|---------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Computational Fluid Dynamics (CFD) | ANSYS Fluent                    | ASME PTC 4.4, EPA Standards, ISO 14001, NFPA 85        | Kiln burner flame aerodynamics, preheater cyclone pressure drop, calciner combustion models, clinker cooler aeration.              |
| Finite Element Analysis (FEA)      | ANSYS Mechanical                | ASME Section VIII Div 1 & 2, Eurocode 3, DIN 18800     | Static structural stress verification, modal & vibration resonance, thermal fatigue life, buckling stability analysis.             |
| Piping Flexibility & Stress        | CAESAR II                       | ASME B31.1 (Power), ASME B31.3 (Process), API 610/617  | Sustained, expansion & occasional stress reports, pump/turbine nozzle loads, spring hanger data sheets, 3D stress isometrics.      |
| Plant Layout & 3D Piping           | AVEVA E3D, Plant 3D, SolidWorks | PIP (Process Industry Practices), ISO 10628, OSHA 1910 | Plant General Arrangement (GA) drawings, 3D equipment models, clash detection reports, piping isometrics, bill of materials (BOM). |
| Structural & Platework             | Tekla Structures, AutoCAD       | AISC 360, ASTM A36/A572, AWS D1.1, IS 800              | Steel fabrication drawings, ductwork layout (process gas, dedusting), transfer chutes (DEM-guided), cyclone hoppers.               |

5 Detailed Functional Requirements (FR)

5.1 Heavy Industrial Domain Solutions Suite (FR-1)

| Req ID | Feature / Module                    | Functional Requirement Specification                                                                                                                                                                                                                                                                                                                                                                                    | Priority       |
|--------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FR-1.1 | Cement Plant Engineering Engine     | <ul style="list-style-type: none"><li>• Deliver complete pyroprocessing engineering: rotary kiln general arrangements, calciner burner ductwork, cyclone preheater strings.</li><li>• Heavy platework design: process gas ducts, dedusting ductwork, baghouse layouts, and clinker cooler transition hoppers.</li><li>• Material handling layout: raw mill, coal mill, and cement grinding circuit plant GAs.</li></ul> | P0 - MANDATORY |
| FR-1.2 | Mining & Minerals Processing Engine | <ul style="list-style-type: none"><li>• Layout design for crushing stations, screening buildings, and beneficiation circuits.</li><li>• Bulk transfer chute design engineered via Discrete Element Modeling (DEM) to mitigate rock impact wear and material degradation.</li><li>• Slurry transport hydraulic calculations and hydrocyclone separation modeling.</li></ul>                                              | P0 - MANDATORY |
| FR-1.3 | Oil & Gas Upstream & Downstream     | <ul style="list-style-type: none"><li>• Complete FEED and detailed engineering packages for upstream skids and downstream refinery units.</li><li>• 3D piping routing in AVEVA E3D / AutoCAD Plant 3D with zero clash tolerance.</li><li>• Equipment mechanical design: ASME Section VIII Div 1/2 pressure vessels and API 650/620 atmospheric storage tanks.</li></ul>                                                 | P0 - MANDATORY |
| FR-1.4 | Multi-Discipline 3D Coordination    | <ul style="list-style-type: none"><li>• Comprehensive clash resolution across piping, civil/structural, electrical cable trays, and HVAC ducting prior to fabrication.</li><li>• Automated extraction of verified fabrication drawings and weight take-offs.</li></ul>                                                                                                                                                  | P1 - HIGH      |

5.2 High-Fidelity Numerical Simulation & Multi-Physics Suite (FR-2)

| Req ID | Feature / Module                        | Functional Requirement Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Priority       |
|--------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FR-2.1 | Computational Fluid Dynamics (CFD)      | <ul style="list-style-type: none"><li>• High-resolution aerodynamic modeling via ANSYS Fluent.</li><li>• Multi-phase reacting flow: Rotary kiln burner flame shape, fuel-air mixing, and calciner NOx reduction dynamics.</li><li>• Pressure drop optimization across preheater cyclone strings and long dedusting runs, targeting ≥ 15% fan power savings.</li><li>• Deliverable: Comprehensive technical report with velocity streamlines, pressure contours, and redesign recommendations within 7-14 days.</li></ul> | P0 - MANDATORY |
| FR-2.2 | Finite Element Analysis (FEA)           | <ul style="list-style-type: none"><li>• Structural verification in ANSYS Mechanical: static linear and non-linear stress, modal resonance, and cyclic thermal fatigue.</li><li>• Dynamic analysis of vibrating equipment foundations (crushers, ball mills, vibrating screens) to prevent structural harmonics.</li><li>• Buckling stability verification for tall process cyclones and storage silos under wind and seismic loading.</li></ul>                                                                          | P0 - MANDATORY |
| FR-2.3 | Piping Stress & Flexibility (CAESAR II) | <ul style="list-style-type: none"><li>• Code compliance qualification per ASME B31.1 and B31.3.</li><li>• Calculation of thermal expansion, seismic loads, and rotating equipment nozzle loading (API 610/617 limits).</li><li>• Generation of spring hanger schedules, pipe support design, and expansion joint specification.</li></ul>                                                                                                                                                                                | P0 - MANDATORY |



5.3 Skilled Engineering Manpower Deputation & Talent Matrix (FR-3)

| Req ID | Feature / Module                  | Functional Requirement Specification                                                                                                                                                                                                                                                                                                                                                                                                   | Priority         |
|--------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| FR-3.1 | 11-Discipline Coverage Matrix     | Provide pre-screened talent across 11 key industrial engineering specialisms:<br>1. Mechanical Equipment • 2. Civil & Structural • 3. Process • 4. Piping • 5. Electrical • 6. Instrumentation & Controls (I&C) • 7. Project Planning (Primavera P6) • 8. Procurement • 9. QA/QC • 10. HSE • 11. Senior CAD/Tekla Drafters.                                                                                                            | P0 - MANDATORY   |
| FR-3.2 | Flexible Deputation Modalities    | <ul style="list-style-type: none"><li>• <b>Onsite Plant Deputation:</b> Direct deployment to client plant sites globally for greenfield construction or turnaround maintenance.</li><li>• <b>Offshore Delivery Center (GDC):</b> Dedicated engineering squads based in Chennai GDC with secure VPN access.</li><li>• <b>Hybrid Augmentation:</b> Combined on-site lead engineer with offshore drafting/modeling back-office.</li></ul> | P0 - MANDATORY   |
| FR-3.3 | Mobilization SLA & Vetting Engine | <ul style="list-style-type: none"><li>• Rapid candidate shortlisting within 48–72 hours; full mobilization within <b>5 to 14 business days</b>.</li><li>• 4-Tier Screening: Technical interview → CAD/FEA test → Code compliance check → Client panel review.</li></ul>                                                                                                                                                                | P1 - OPERATIONAL |

5.4 Digital Web Platform & PWA Offline Infrastructure (FR-4)

| Req ID | Feature / Module               | Functional Requirement Specification                                                                                                                                                                                                                                                        | Priority       |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| FR-4.1 | Zero-Framework Core Platform   | <ul style="list-style-type: none"><li>• Built with Semantic HTML5, Modular CSS3 Custom Properties, and ES6+ JavaScript.</li><li>• Zero heavy framework runtime (no React/Vue hydration overhead), achieving sub-second Time-to-First-Byte (TTFB) and zero Layout Shift (CLS = 0).</li></ul> | P0 - MANDATORY |
| FR-4.2 | Service Worker Caching (sw.js) | <ul style="list-style-type: none"><li>• Dual-strategy caching: <i>Cache-First</i> for static assets (CSS, JS, WebP/PNG logos, fonts) and <i>Stale-While-Revalidate</i> for core HTML.</li><li>• Enables complete offline resilience and instantaneous sub-50ms repeat page loads.</li></ul> | P0 - MANDATORY |
| FR-4.3 | Consultation & Lead Engine     | <ul style="list-style-type: none"><li>• Interactive modal consultation form with client-side field validation, sector categorization, and CSRF protection.</li><li>• Instant mobile fallback bridge to direct WhatsApp technical advisory (+91 76048 52835).</li></ul>                      | P1 - HIGH      |

5.5 Machine Intelligence & Autonomous Discovery Endpoints (FR-5)

| Req ID | Feature / Module                | Functional Requirement Specification                                                                                                                                                                                                                                                                        | Priority        |
|--------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| FR-5.1 | LLM-Ready Protocols (/llms.txt) | <ul style="list-style-type: none"><li>• Implement compliant /llms.txt summary and /llms-full.txt endpoints delivering deterministic, plain-text markdown specifications for LLMs and autonomous procurement crawlers.</li><li>• Eliminates HTML parsing errors and enables automated RFQ scoring.</li></ul> | P1 - INNOVATION |
| FR-5.2 | Schema.org Knowledge Graph      | <ul style="list-style-type: none"><li>• Rich JSON-LD microdata on every page specifying Organization, WebSite, FAQPage, and Person schemas for Google Knowledge Graph and Rich Answer Snippets.</li></ul>                                                                                                   | P1 - INNOVATION |



## 6 Non-Functional Requirements (NFR)

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The engineering deliverables and digital infrastructure must satisfy strict operational, security, and quality constraints:

### 6.1 Performance & Latency Requirements (NFR-1)

- **Digital Platform:** Largest Contentful Paint (LCP)  $\leq 1.2s$  on mobile 4G; First Input Delay (FID) / INP  $\leq 50ms$ ; Cumulative Layout Shift (CLS) = 0.00.
- **PWA Repeat Load Latency:** Assets served from Service Worker cache must respond in  $\leq 50ms$  (measured locally at ~25-35ms).
- **Simulation Report Turnaround:** Standard CFD/FEA diagnostic package delivered within **7 to 21 business days** from geometry freeze.

### 6.2 Reliability, Availability & Resilience (NFR-2)

- **Web Availability:** 99.95% uptime hosted on global edge CDN with automated failover.
- **Offline Functionality:** Precached core pages (`index.html`, `Contact.html`, `services.md`) must render fully even when network connectivity drops to zero.
- **Drawing Integrity:** All CAD files and calculations backed up incrementally daily with offsite encrypted replication.

### 6.3 Information Security, IP & Confidentiality (NFR-3)

- **Non-Disclosure Agreement (NDA):** Executed prior to exchanging proprietary CAD files, plant diagrams, or operational data.
- **Intellectual Property (IP) Transfer:** 100% of generated CAD drawings, simulation models, and calculation sheets transfer to the client upon final milestone settlement.
- **Data Governance:** Full compliance with ISO 27001 data isolation practices; client drawings stored in dedicated project repositories with role-based access control (RBAC).

### 6.4 Accessibility & Cross-Platform Compatibility (NFR-4)

- **Accessibility:** Adherence to WCAG 2.1 Level AA guidelines, including minimum 4.5:1 color contrast ratios, full keyboard navigability, and ARIA labels.
- **Viewport Adaptability:** Flawless rendering from 320px mobile displays up to 4K ultra-wide engineering monitors.
- **Browser Matrix:** Full parity across Microsoft Edge, Google Chrome, Mozilla Firefox, Apple Safari (iOS/macOS), and standard embedded WebViews.

### 6.5 Code Quality, Standards & Maintainability (NFR-5)

- **Engineering Code Alignment:** Mandatory compliance with ASME Boiler & Pressure Vessel Code (BPVC), ASME B31.1/B31.3, API 650/620, ASTM, DIN, and ISO 9001.
- **Software Maintainability:** Strict separation of concerns, zero inline styling in production templates, semantic markup, and zero external runtime npm dependencies for the client frontend.

## 7 Commercial Engagement Models & Milestone Stage-Gate Governance

### 7.1 Commercial Engagement Models Matrix

| Engagement Model                  | Commercial Structure                           | Scope & Modality                                                                                                 | Key Terms & Invoicing Gates                                                                                         |
|-----------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Model A: Manpower Deputation      | Time & Materials (T&M) / Monthly Retainer      | Onsite plant assignment, Chennai Offshore GDC, or Hybrid augmentation across 11 disciplines.                     | Short-term (1–6 mo) or Long-term (6–24+ mo). Mobilization in 5–14 days. Monthly invoicing with verified timesheets. |
| Model B: Fixed-Scope Turnkey EPCM | Milestone-Based Fixed Price Contract           | Complete plant GA drawings, FEED packages, heavy platework fabrication drawings, and BOMs.                       | Stage-Gate Tranches: 30% Advance • 30% at FEED/60% • 30% at Detailed/90% • 10% upon Final IFC release.              |
| Model C: Numerical Simulation     | Per-Study Lump Sum / Master Services Agreement | ANSYS Fluent CFD, ANSYS Mechanical FEA, and CAESAR II piping stress evaluations.                                 | Diagnostic SLA 7–21 business days. Invoiced 50% upon CAD geometry freeze, 50% upon formal report delivery.          |
| Model D: Technical Recruitment    | Permanent Placement / Retained Search Fee      | Executive search for licensed site directors, plant heads, lead process engineers, and chief technical officers. | Success fee tied to candidate joining date with a 90-day replacement guarantee.                                     |

### 7.2 The 4-Stage Milestone Engineering Gateway

Fixed-scope engineering deliverables strictly follow the **Stage-Gate Review Process** to eliminate scope creep and ensure zero rework:



**Engineering Revision Policy:** Standard fixed-scope contracts include **two complete rounds of client design comments** and iteration prior to Gate 4 IFC sign-off. Any subsequent changes requiring structural re-modeling or re-meshing are managed via formal Engineering Change Orders (ECO).

8 Release Roadmap, Phasing & Resource Allocation

8.1 Multi-Phase Execution & Feature Roadmap

| Phase                  | Timeline          | Key Capabilities & Engineering Milestones                                                                                                                                                                                                   | Status      |
|------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Phase 1: Baseline      | Q1 – Q3 2025      | <ul style="list-style-type: none"><li>Cement &amp; Mining plant engineering consulting launch.</li><li>11-Discipline manpower deputation mobilization.</li><li>Corporate web platform launch and domain registration.</li></ul>             | COMPLETED   |
| Phase 2: Platform Core | Q4 2025 – Q2 2026 | <ul style="list-style-type: none"><li>Full Service Worker (PWA) dual-cache deployment (sw.js).</li><li>ANSYS Fluent &amp; Mechanical multi-physics standardization.</li><li>Schema.org JSON-LD graph &amp; /IImS.txt integration.</li></ul> | COMPLETED   |
| Phase 3: Client Portal | Q3 – Q4 2026      | <ul style="list-style-type: none"><li>Interactive client RFQ configurator with 3D model viewer.</li><li>Client Deliverable Tracking Portal with Stage-Gate approval logs.</li><li>Automated CAESAR II stress report exporter.</li></ul>     | IN PROGRESS |
| Phase 4: AI Estimator  | Q1 – Q2 2027      | <ul style="list-style-type: none"><li>AI-assisted engineering proposal and steel weight estimation.</li><li>Automated duct aerodynamic pressure-drop preliminary sizing.</li><li>Global talent mobilization matching algorithm.</li></ul>   | PLANNED     |

8.2 Multi-Disciplinary RACI Governance Matrix

| Operational / Project Activity                | Exec Sponsor | Senior PM   | Lead Eng    | CFD/FEA     | QA/QC       | Digital PM  |
|-----------------------------------------------|--------------|-------------|-------------|-------------|-------------|-------------|
| Project Scope & Commercial Proposal           | ACCOUNTABLE  | RESPONSIBLE | CONSULTED   | INFORMED    | INFORMED    | INFORMED    |
| Plant Layout & GA Drawing Packages            | INFORMED     | ACCOUNTABLE | RESPONSIBLE | CONSULTED   | CONSULTED   | INFORMED    |
| Numerical Simulation (CFD / FEA / Stress)     | INFORMED     | CONSULTED   | ACCOUNTABLE | RESPONSIBLE | CONSULTED   | INFORMED    |
| Engineering Manpower Screening & Mobilization | ACCOUNTABLE  | RESPONSIBLE | RESPONSIBLE | INFORMED    | CONSULTED   | INFORMED    |
| Stage 4 Final IFC Quality Audit & Handover    | INFORMED     | CONSULTED   | RESPONSIBLE | CONSULTED   | ACCOUNTABLE | INFORMED    |
| Digital Web Platform, PWA & AI Endpoints      | INFORMED     | ACCOUNTABLE | INFORMED    | INFORMED    | INFORMED    | RESPONSIBLE |

9 Risk Management, Mitigation & Contingency Matrix

Comprehensive risk evaluation across computational, resourcing, compliance, and technological domains:

| Risk ID | Identified Risk Event                                                                                                                   | Severity / Prob. | Preventative Mitigation Strategy                                                                                                      | Contingency Action                                                                            |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| RSK-01  | <b>Simulation Mesh Divergence</b><br>Complex 3D geometries (e.g. cyclonic combustor) fail to converge within expected iterations.       | MEDIUM • LOW     | Standardized grid convergence index (GCI) protocol; polyhedral meshing in ANSYS Fluent; pre-run CAD defeaturing.                      | Deploy dedicated HPC cloud nodes with double precision solver; 48-hr SLA extension alert.     |
| RSK-02  | <b>Deputation Visa / Travel Delays</b><br>International client plant sites face visa delays for on-site deputed personnel.              | HIGH • MEDIUM    | Maintain pre-cleared roster of internationally mobile engineers with active multi-entry business visas; initiate mobilization on LOI. | Immediate interim mobilization of Chennai Offshore GDC team with remote plant telemetry.      |
| RSK-03  | <b>Late-Stage Client Scope Creep</b><br>Client requests extensive ductwork re-routing after Gate 2 FEED freeze.                         | HIGH • MEDIUM    | Formal Stage-Gate freeze sign-off; standard contract stipulating two review cycles max before IFC.                                    | Issue Engineering Change Order (ECO) with transparent schedule impact and cost addendum.      |
| RSK-04  | <b>Code Non-Conformance in Fabrication</b><br>Field welding contractor discovers nozzle reinforcement deficit under ASME B31.3.         | CRITICAL • LOW   | 100% dual-engineer checking routine; independent QA/QC verification sign-off prior to Gate 4 drawing stamping.                        | 24/7 on-call engineering advisory desk to provide approved structural repads within 12 hours. |
| RSK-05  | <b>Browser Cache Staling (PWA)</b><br>Clients visiting web portal see stale pricing or services due to aggressive Service Worker cache. | LOW • LOW        | Cache-busting version keys (vqube-perf-v3); immediate cache purging on activation event in sw.js.                                     | Automated version check script alerting client when newer bundle is ready to reload.          |

**Enterprise Risk Governance Protocol:** The PMO maintains a live, bi-weekly risk register. Any issue crossing Severity Level 4 triggers an automatic Emergency Engineering Review chaired by Vadivelu Dhamodharan (Head — Engineering & Business Solutions) with resolution mandatory within 24 business hours.

10 Quality Verification, Definition of Done & Senior Sign-Off

10.1 Definition of Ready (DoR) & Definition of Done (DoD)

**Definition of Ready (DoR) for Engineering:**

- Executed Client Contract & Mutual Non-Disclosure Agreement (NDA).
- Approved Basis of Design (BOD) including operating temperatures, pressures, material specs, and seismic/wind parameters.
- Approved 3D equipment boundary envelope or vendor general arrangements.
- Designated client technical counterpart for Stage-Gate review sign-offs.

**Definition of Done (DoD) for IFC Deliverables:**

- Zero outstanding clash detections in multi-discipline 3D model.
- ANSYS FEA/CFD and CAESAR II stress analysis reports peer-reviewed and signed off.
- Full code compliance verified against ASME, ASTM, DIN, ISO, and API standards.
- 100% of CAD drawings stamped "ISSUED FOR CONSTRUCTION (IFC)" with complete BOMs and steel take-offs.
- All IP transferred to client; archive snapshot stored in secure repository.

10.2 Final Quality Verification Checklist

| Verification Dimension             | Verification Standard / Acceptance Criteria                                    | Compliance Audit   |
|------------------------------------|--------------------------------------------------------------------------------|--------------------|
| Boilers & Pressure Vessels         | ASME BPVC Section VIII Div 1 & 2; wall thickness, nozzle necks & weld joints   | 100% CODE VERIFIED |
| Process Piping & Thermal Expansion | ASME B31.3 / B31.1 code compliance via CAESAR II finite element flexibility    | 100% CODE VERIFIED |
| Computational Fluid Dynamics       | Grid convergence index (GCI) < 3%; mass & energy residual imbalance < 0.1%     | 100% CODE VERIFIED |
| Manpower Deputation Quality        | All deployed engineers certified; client sign-off on 4-tier screening matrix   | 100% CODE VERIFIED |
| Web Platform & PWA Core Vitals     | 100/100 Lighthouse performance; sub-50ms repeat load via sw.js; valid /lms.txt | 100% CODE VERIFIED |

10.3 Senior Executive Approval & Authorization Block

By signing below, the undersigned executive authorities formally baseline and approve this **Product Requirements Document (PRD-VQT-2026-V2.4)** as the definitive technical and operational benchmark for VQube Technologies.

| Designation                             | Executive Name              | Signature / Seal             | Date of Approval  |
|-----------------------------------------|-----------------------------|------------------------------|-------------------|
| Senior Project & Product Manager        | Lead Author & Delivery Head | [AUTHORIZED • DIGITAL PMO]   | 27 September 2026 |
| Head — Engineering & Business Solutions | Vadivelu Dhamodharan        | [APPROVED • EXECUTIVE BOARD] | 27 September 2026 |
| Quality Assurance & Compliance Director | Lead Quality Inspector      | [VERIFIED • ISO/ASME QA]     | 27 September 2026 |

